

Understanding NVIDIA Cosmos

A beginner's guide to the four generations — Cosmos 1, 2, 2.5 and 3 — and what actually changes between them.

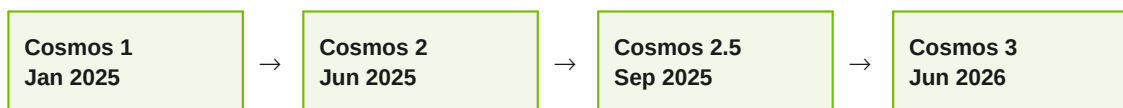
First: what is a “World Foundation Model”?

You have probably heard of a **language** model (like the one writing this): it learns from text and predicts the next word. A **World Foundation Model (WFM)** does the same trick, but for the *physical world*. It watches enormous amounts of video and learns to predict **what happens next** — how a ball bounces, how a hand grips a cup, how a car moves through a street.

NVIDIA **Cosmos** is a family of these models, built for “**Physical AI**” — robots and self-driving cars. Instead of testing a robot a million times in the real world (slow, expensive, dangerous), engineers let Cosmos **imagine** realistic video of what would happen, and train on that.

Analogy: think of Cosmos as a “dream engine” for machines. You describe a scene, and it dreams a physically-believable video of how that scene plays out. Each new generation dreams better, faster, and about more things (now even sound and physical actions).

The story in one line



Plain-English summary of the journey: Cosmos started as a big toolbox with **two** different engines, then **simplified and unified** into one engine, then made that engine **smarter and longer**, and finally became an **all-in-one model that also reasons, hears, and acts**.

Side-by-side comparison

What	Cosmos 1	Cosmos 2	Cosmos 2.5	Cosmos 3
Released	Jan 2025	Jun 2025	Sep 2025	Jun 2026
Main idea	Toolbox with TWO engines (diffusion + GPT-style)	Refined the diffusion engine; faster, cleaner	ONE unified engine (flow-based)	All-in-one “omnimodel” that reasons + generates + acts
What goes in	Text or a starting image/video	Text or image/video	Text, image, or video (one model)	Text, image, video, sound, AND actions
What comes out	Future video	Future video (480p/704p)	Future video up to 30s, multi-camera	Video, actions, audio — with reasoning
How it reads text	T5 text encoder	T5-style encoder	Cosmos-Reason1 (a smart vision-language model)	Built-in reasoning transformer
Sizes offered	7B/14B (diffusion), 4B/12B (GPT)	2B and 14B	2B and 14B	Super / Nano / Edge
One-word vibe	Kitchen sink	Polish	Unify	Leap

Now, generation by generation

Cosmos 1 — “The Toolbox” (Jan 2025)

This was the first release, and it shipped **two completely different ways** to dream up video:

- **Diffusion engine** (sizes 7B & 14B): starts from random noise and gradually “cleans it up” into a video. Slower, but the picture quality is excellent.
- **Autoregressive / GPT engine** (sizes 4B & 12B): predicts the video one chunk at a time, like typing one word after another. Faster, aimed at real-time, but rougher — so it needs a clean-up step afterwards.

It also split tasks into **separate models**: one for text→video, one for image→video, one for video→video. Powerful, but a lot of moving parts.

Beginner takeaway: Cosmos 1 gave you many tools and asked you to pick the right one. Great flexibility, but you had to know what you were doing.

Cosmos 2 — “The Polish” (Jun 2025)

Instead of adding more, Cosmos 2 **focused**. It doubled down on the diffusion engine and made it **faster, sharper, and more obedient** to your prompt — with fewer “hallucinations” (made-up nonsense). It came in a small **2B** (fast) and a big **14B** (high quality) version, at 480p/704p resolution.

Beginner takeaway: same idea as Cosmos 1, just done much better and with fewer mistakes.

Cosmos 2.5 — “The Unification” (Sep 2025)

The big simplification. The three separate task-models from before were **merged into one** model that accepts text, image, OR video. It switched to a **flow-based** method (a cleaner cousin of diffusion) and, importantly, started using **Cosmos-Reason1** — a model that actually *understands* language and the physical world — to read your prompt. Results: video up to **30 seconds** long and **multiple synchronized camera angles**.

Beginner takeaway: one smart model instead of three, with a much better “ear” for what you asked, and longer videos.

Cosmos 3 — “The Leap” (Jun 2026)

Not just a better video maker — a different kind of model. Cosmos 3 is an **“omnimodel”**: one system that natively handles **text, images, video, sound, and physical actions**. It uses a **mixture-of-transformers** design that pairs a **reasoning** part (thinks about how objects interact) with a **generation** part (creates the video and the action). In other words, it **thinks before it dreams**. It comes as **Super** (max accuracy), **Nano** (answers in a split second), and **Edge** (runs locally, coming soon).

Beginner takeaway: earlier versions were “dream engines.” Cosmos 3 adds a brain and senses — it reasons, sees, hears, and can decide what a robot should physically do.

If you remember only one thing...

- **Cosmos 1** = many tools, two different engines. (*Flexible but complex.*)
- **Cosmos 2** = the best engine, polished. (*Better & fewer mistakes.*)
- **Cosmos 2.5** = one unified, smarter engine, longer video. (*Simpler & sharper.*)
- **Cosmos 3** = an all-in-one model that reasons, sees, hears, and acts. (*A new category.*)

Sources: NVIDIA Cosmos World Foundation Model Platform paper (arXiv:2501.03575); NVIDIA newsroom & developer blogs on Cosmos Predict-2, Predict-2.5, and Cosmos 3; Hugging Face NVIDIA blog on Cosmos Predict/Transfer 2.5. Generated as an educational summary.